

TECHNICAL BRIEF

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Key Points:

- A visualization program for the Yin-Yang grid system is developed
- The program is coded in Fortran 90 with hardware-rendering using GPU
- Yin-Yang grid users may develop own visualization tools based on this code

Supporting Information:

- Short description of tar files in Supporting Information
- sv4 source codes
- sample data files

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A Fortran visualization program for spherical data on a Yin-Yang grid

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Abstract A Fortran 90 program to visualize data on the Yin-Yang grid system is developed. The purpose of this study is to provide simulation researchers with a source code as a starting point of their own custom-made visualization tools. A basic but sufficiently diverse set of visualization methods are implemented using a Fortran 90 binding for OpenGL for scalar and vector fields defined or simulated on the Yin-Yang grid.

1. Introduction

The Yin-Yang grid (YYG), which was proposed in *G-cubed* about a decade ago [Kageyama and Sato, 2004], is a spherical grid system composed of two congruent grids—Yin and Yang—with partial overlap (Figure 1). The boundary data on each component grid are set by mutual interpolations following the standard overset grid methodology [Chesshire and Henshaw, 1990].

The use of spherical geometry simulations using YYG as a base grid system is increasing in various fields. Examples in geo-physics and planetary physics include simulations of the geodynamo, which was the original motivation of the development of the YYG [Kageyama et al., 2008; Miyagoshi et al., 2010, 2011], mantle convection [Kameyama et al., 2008; Tackley, 2008; Nakagawa and Tackley, 2008, 2011; Deschamps et al., 2012], ionosphere/magnetosphere coupling [Sawada and Ogino, 2005], plate dynamics of the Earth [Coltice et al., 2012], and Martian dichotomy [Reese et al., 2010, 2011]. General circulation models of the atmosphere and/or ocean using YYG have been studied in Japan [Baba et al., 2010], Canada [Qaddouri, 2011; Qaddouri et al., 2012], England [Zerroukat and Allen, 2012], and China [Li et al., 2012]. The YYG system has also been applied actively in solar physics and astrophysics. The static [Feng et al., 2012] and dynamic magnetic fields [Shiota et al., 2010] of the solar corona and the solar wind [Feng et al., 2011] as well as solar convection and dynamo [Masada et al., 2013; Hotta et al., 2014], have been modeled and simulated on YYG. At a considerably larger spatial scale, simulations of supernovae [Wongwathanarat et al., 2010; Müller et al., 2012; Warren and Blondin, 2013] and accretions [Blondin and Raymer, 2012] have also been performed on YYG.

A merit of YYG compared with other spherical grid systems such as the cubed sphere [Ronchi et al., 1996] is that the component grids of YYG are nothing but a part of the standard latitude-longitude grid system defined on the spherical polar coordinates whose metric and differential operators are familiar to simulation researchers. As a consequence, visualization of data defined on the YYG is relatively easy since general purpose and freely available visualization software such as ParaView, VisIt, Vistool, VTK, and Vapor, as well as commercially available software products can be straightforwardly used. However, there continue to be compelling reasons for developing specialized visualization software customized for YYG. One reason is the steep learning curve required for a simulation researcher to become skilled in a general purpose visualization program because of the numerous elements to be understood in such rich visualization packages. Another reason is that it is common for simulation researchers to customize the source code of visualization tools to fit their original visualization purposes; however, modern general purpose software aims at being complete, preventing it from being handy. It is generally impractical to expect a geophysical simulation researcher to grasp the entire structure of open-source visualization software to specify a proper location to change the source code for their own visualization tasks. We believe that simulation researchers using YYG need basic, easy-to-read, and easy-to-change visualization software that is complementary to the rich and advanced general purpose software. The purpose of this *Technical Brief* is to report the development of such a minimal set of sample source code that can be used as a starting point or template by YYG users.

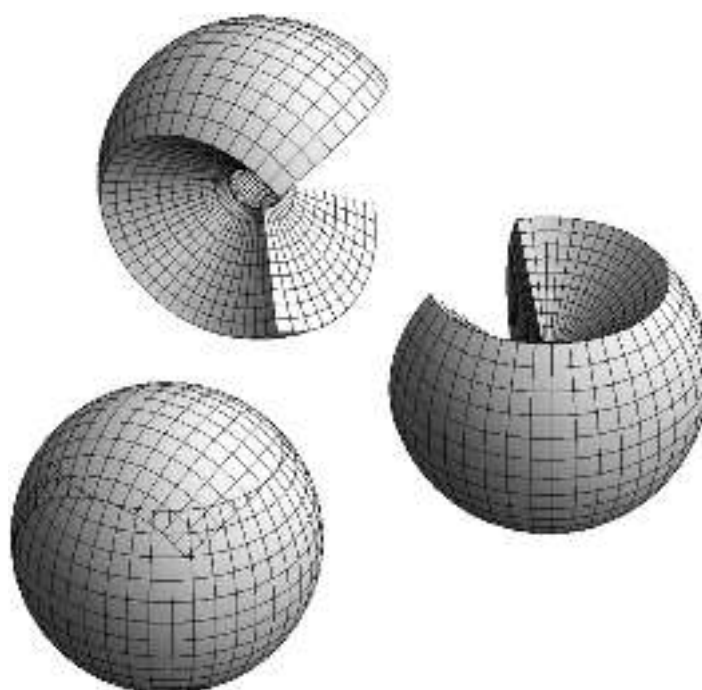


Figure 1. (right) Yin-component, (top left) Yang-component, and (bottom left) Yin-Yang grid.

A couple of visualization software packages for YYG have been developed. Greensky *et al.* [2008] designed a visualization system for mantle convection simulations. This is an integrated visualization system composed of hardware and software that generates volume rendering images from various kinds of mantle convection simulations. Further, Ohno and Kageyama [2009] developed visualization software, named Armada, for data defined on YYG, and which is a software-rendering visualization program. Armada is written in C++ and parallelized by OpenMP and MPI. It may run on supercomputers since it does not need OpenGL API and graphics hardware (GPU).

The visualization software developed in this study, SV4, is written in Fortran 90 language. In con-

trast to Armada, SV4 invokes hardware-rendering commands to GPU through the OpenGL. In general, if GPU based hardware-rendering is agreed on, there is no strong reason to stick to a particular computer language so long as it provides a proper interface, or binding, to OpenGL. However, the main purpose of this study is to provide customizable visualization software for YYG to simulation researchers. The choice of language matters in this regard because we believe that there are more Fortran users than those of C++, at least in the geophysical simulation community.

2. SV4 Program

Since our main objective is to provide a starting point for custom-made visualization tools for each simulation researcher, we implemented a minimal number of basic visualization methods in SV4, while trying to maintain as broad a range of visualization algorithms as possible. The implemented methods are isosurface, color plot on cross sections, and streamlines, as described below.

To use OpenGL with Fortran 90 language, we use f90gl [Mitchell, 1999], which constitutes the official Fortran 90 bindings for OpenGL, and is approved by the OpenGL Architecture Review Board. f90gl is a public domain software.

SV4 can read multiple scalar and vector fields. To keep the source code's simplicity and portability, the input data of SV4 are the raw binary, written by *write* function with *unformatted* format. If users need more sophisticated numerical data format such as netCDF or HDF, it would be easy to incorporate them into this template source code.

The source code files of SV4 are provided in supporting information. Download and extract **sv4.tar.gz**. You will find **src** and **data** directories. The source codes are in **src**. After a proper path setting to f90gl library (see **README.txt**), a make command will simply generates SV4. If you want to try SV4 with a sample data set, download **sample_data.place_these_files_in_data_folder.tar** in supporting information and place the extracted data files in **data** directory.

2.1. Isosurface

Isosurface is constructed using the classical marching cubes algorithm proposed by Lorensen and Cline [1987]. An elemental cube is defined as a small rectangular region for both the Yin and Yang-component grids.

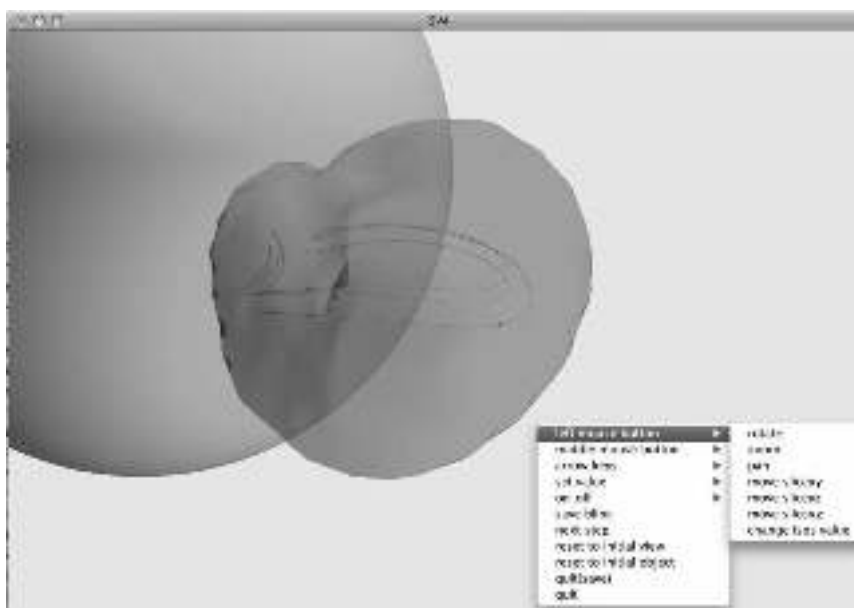


Figure 2. Isosurface of the magnetic field amplitude visualized by SV4.

We apply the marching cubes algorithm on both the Yin and Yang-component grids, with no special attention to the overlapped region. In other words, for a level of the target scalar field, two surfaces are independently constructed in the Yin and Yang grids in the overlapped region. The two isosurfaces are, however, located in exactly the same position, because, in the marching cubes algorithm, each polygon of an isosurface is constructed by the interpolation of the scalar field from the background grid, and the field data in the overlapped region is unique, at least to the accuracy of interpolation, because of the mutual interpolation between Yin and Yang grids. We have visually confirmed that this redundant construction of the isosurface does not give rise to any artifact after rendering.

Figure 2 shows a snapshot of an isosurface of the magnetic field amplitude of a solar coronal simulation. To visualize the inner part of the surface, the isosurface can be made transparent as shown in this figure. The target scalar field to be visualized by the isosurface is specified through a graphical user interface (GUI) constructed using GLUT API (see the plot windows in Figure 2). f90gl provides the interface to not only the raw OpenGL functions, but also to GLUT functions.

2.2. Color Plot

Another visualization method for scalar fields implemented in SV4 is color plot on cross sections. The cross sections are specified either by Cartesian coordinates or spherical coordinates. When the x axis is specified, for example, a color plot of a specified scalar field on a plane perpendicular to the x axis is shown. The scalar field or position of the cross section can be controlled through the GUI. Figure 3 shows an example of the constant z plane (on the equator) of a solar coronal simulation.

When the r axis is specified, a sphere of a specified radius appears on which the color plot of a scalar field chosen through another menu is shown. Figure 4 shows a gray scale plot of the radial component of a magnetic field on the solar surface. A pair of strong positive and negative magnetic fluxes are observed.

2.3. Streamline

As a basic visualization method for vector fields, we implemented the streamline method. A streamline $\mathbf{s}(\tau)$ represented by a parameter τ for a vector field $\mathbf{u}(\mathbf{x})$, starting from a seed point $\mathbf{s}(0)$, is defined by the differential equation $d\mathbf{s}(\tau)/d\tau = \mathbf{u}(\mathbf{s}(\tau))$. This equation is numerically integrated by a sixth-order Runge-Kutta

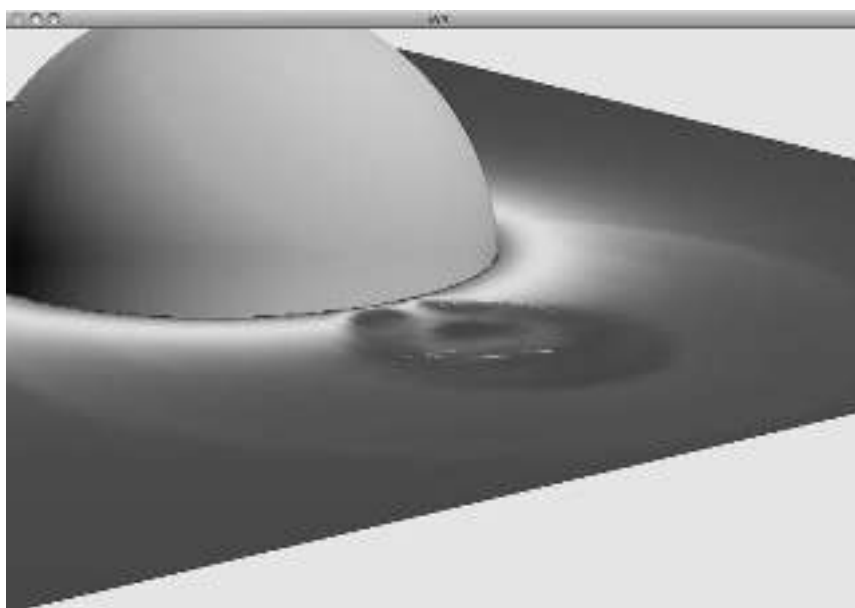


Figure 3. Color plot of a magnetic component in the equatorial plane.

method in SV4. Third-order interpolation is applied to intergrid points. A streamline of the magnetic field $\mathbf{B}(\mathbf{x})$ is nothing but the line of force of the magnetic field.

When the point $\mathbf{s}$ is located in, for example, Yin-grid, the vector value $\mathbf{u}(\mathbf{s})$ is interpolated from data on Yin-grid. Even if a traced streamline runs into the overlapped region, we keep the spatial interpolation only from Yin-grid because the target vector field in the overlapped region is unique due to the mutual interpolation. When the point $\mathbf{s}$ goes out of Yin-grid, the spatial interpolation is switched to be taken from Yang-grid.

In SV4, a seed point $\mathbf{s}(0)$ can be specified, which is denoted by a small red ball in Figure 5, through the GUI with the mouse. The red ball can be picked up and dragged in any direction. Numerical integration of a

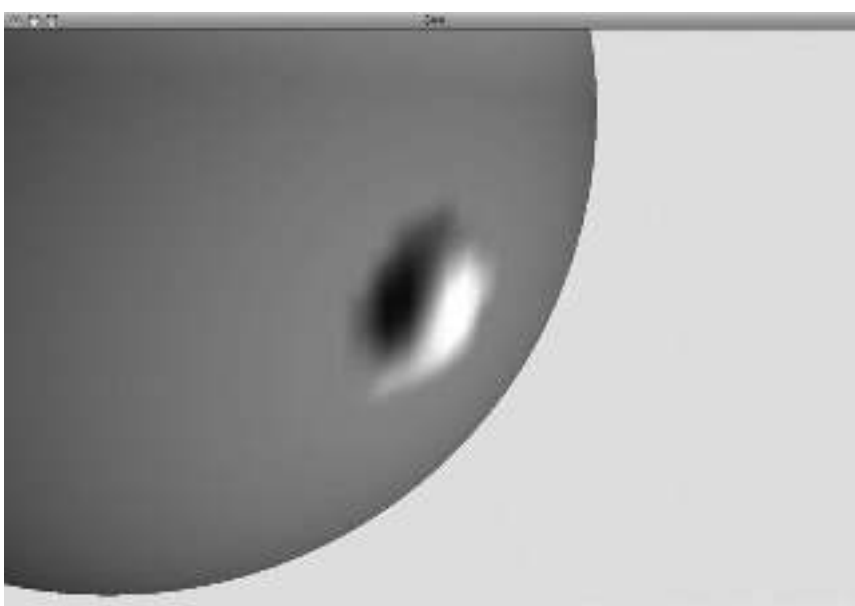


Figure 4. A radial cross section of a magnetic component.

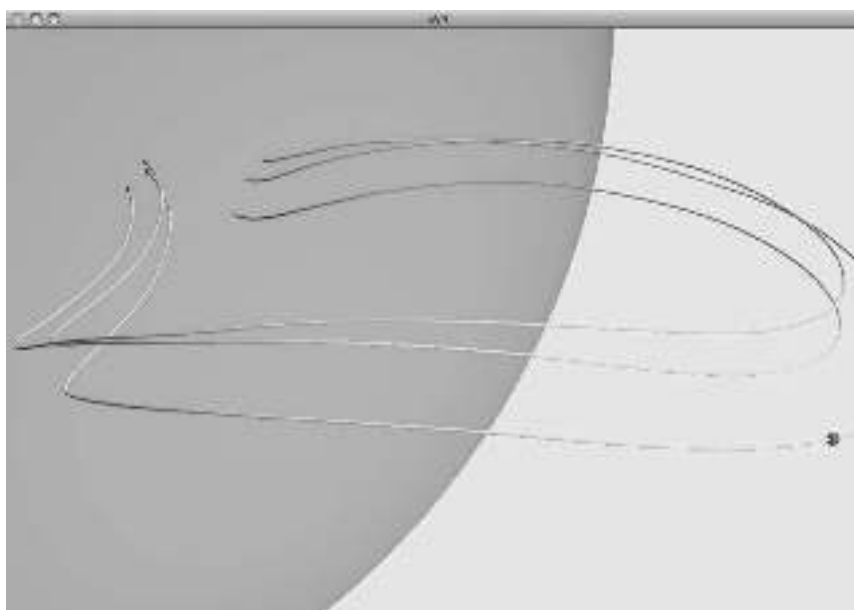


Figure 5. Magnetic field lines drawn by SV4. The seeding point is denoted by a red ball which can be moved by the mouse.

streamline is performed in real time when a drag motion is carried out. A drastic change in the traced streamline can be observed when the line runs through a small region including a bifurcation point as the seeding point is moved with the mouse.

2.4. User Interface

In addition to the basic visualization methods and the related interface described thus far, SV4 includes basic functions such as zooming, rotating, and translating objects. They are applied interactively by intuitive control by the mouse.

The minimal visualization methods and interfaces implemented in SV4 are intentionally kept simple and minimum to make it for a simulation researcher, who is an expert in number crunching with Fortran but may not necessarily be familiar with modern computer graphics and visualization techniques, easy to add new visualization methods, user interface, and new statistical data processing.

3. Summary

We developed a visualization program, SV4, in Fortran 90 for visualization of data defined or calculated on the Yin-Yang grid. By making use of the f90gl API, we can use OpenGL directly from Fortran 90. The main purpose of the development of SV4 is to provide a template source code for simulations researchers who wish to develop their own visualization tools for YYG data.

We also developed the Armada [Ohno and Kageyama, 2009] program for visualization of data on YYG. Armada is a software-rendering program that needs no OpenGL and no GPU. More visualization methods are implemented in Armada than in SV4: volume rendering, colored cross section, isosurface, vector glyph, and stream-tubes. These are all rendered by the ray-casting algorithm without invoking GPU acceleration. Since Armada, which is written in C++, is parallelized by MPI and OpenMP, it can run on massively parallel supercomputers.

On the other hand, the software SV4 reported in this paper, which is coded in Fortran 90, is intended to be a hardware-rendering software, and is supposed to run on a PC or a graphic workstation with GPU. The implemented visualization methods are kept a minimum (streamline, colored cross section, and isosurface) with a simple GUI. The main purpose of SV4 is to provide a starting point of private visualization software for each simulation researcher in each field.

We regard Armada and SV4 as playing a complementary role in serving the expanding community in the field of geophysics who use or intend to YYG as a base grid system.

Acknowledgments

The source codes of SV4 and sample data are available as supporting information. The sample data shown in the figures were provided by D. Shiota and K. Kusano. The authors thank A. Tomiyama for technical assistance in the development and test of SV4 program. This work was supported by grant-in-aid for Scientific Research (KAKENHI) 23340128.

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